Transaction Data Analytics Report

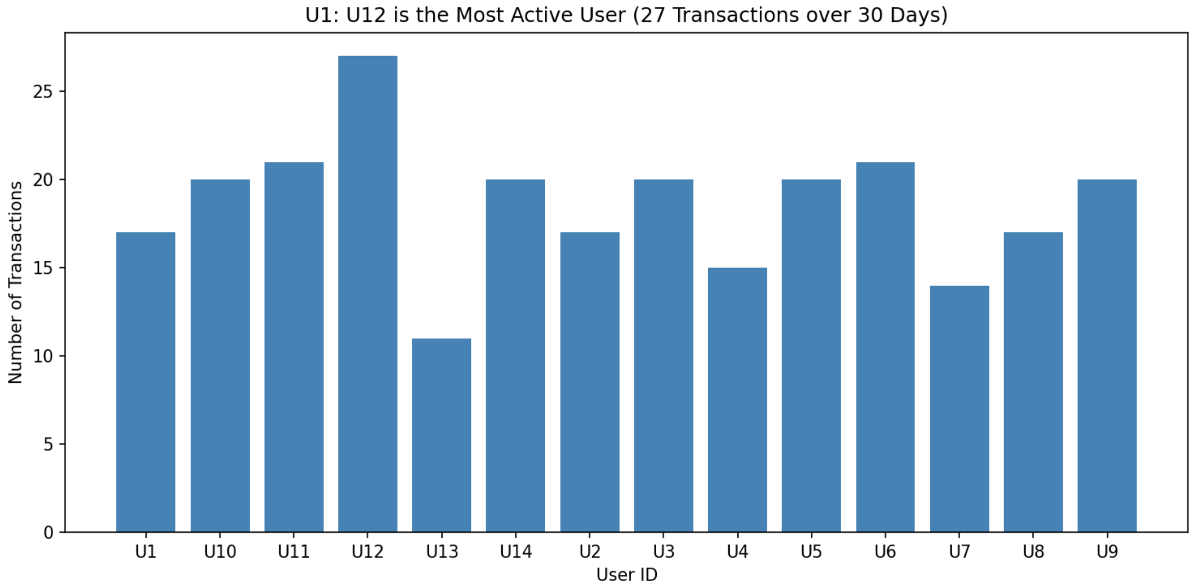
S20230030407 | IDHV | Assignment 2

This report presents insights from 30 days of retail transaction data. The data covers 260 transactions across 14 users, 26 items, and 30 days, generating a total revenue of $466,780.93. The analysis is divided into three sections: user behavior, item performance, and transaction patterns.

1. User Analysis

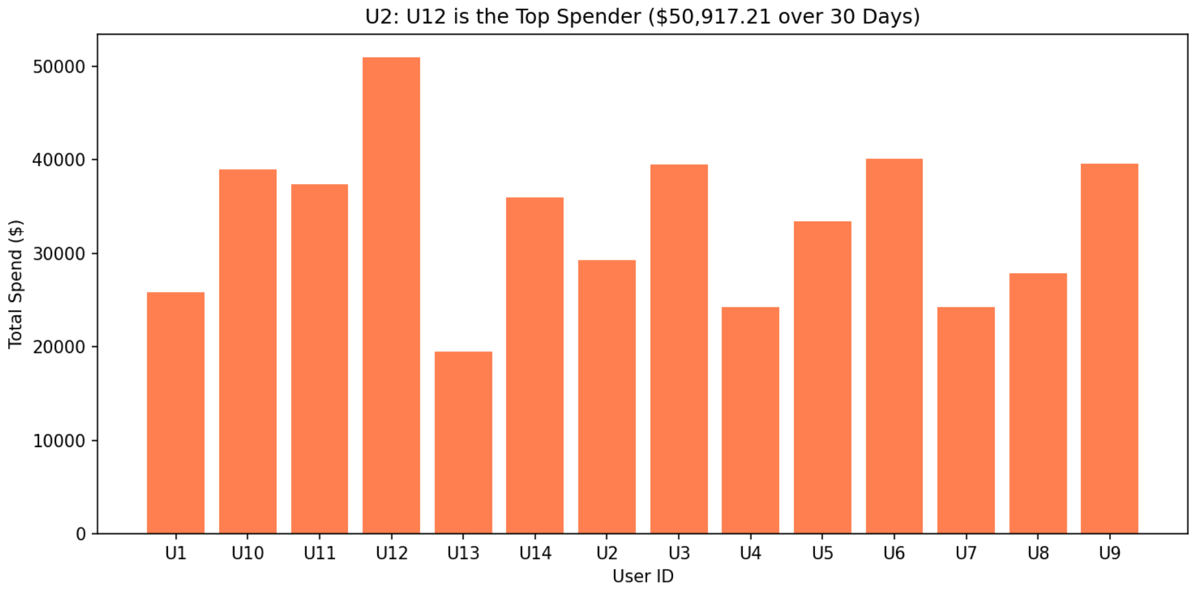
This section examines how different users engaged with the platform over 30 days, looking at frequency, spending, loyalty, and shopping habits.

U1. Transaction Frequency per User



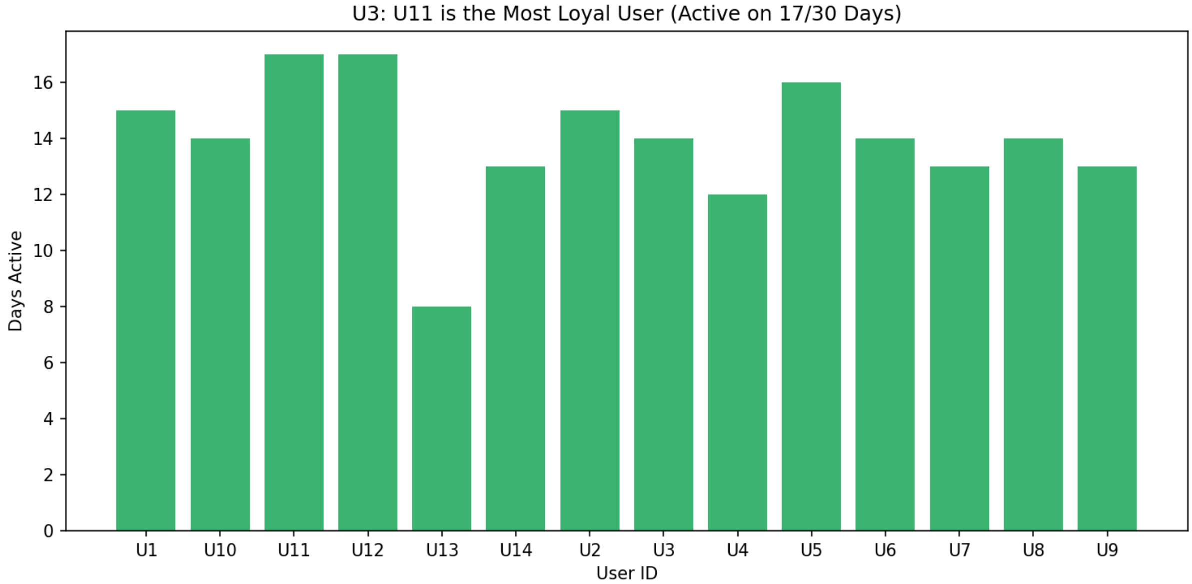
U12 made 27 transactions over 30 days, more than any other user. U13 was the least active with only 11 transactions. Most users fell in the 17 to 20 range, suggesting fairly consistent engagement across the platform. The gap between the most and least active users is notable and points to significant variation in how often different users return.

U2. Total Spend per User



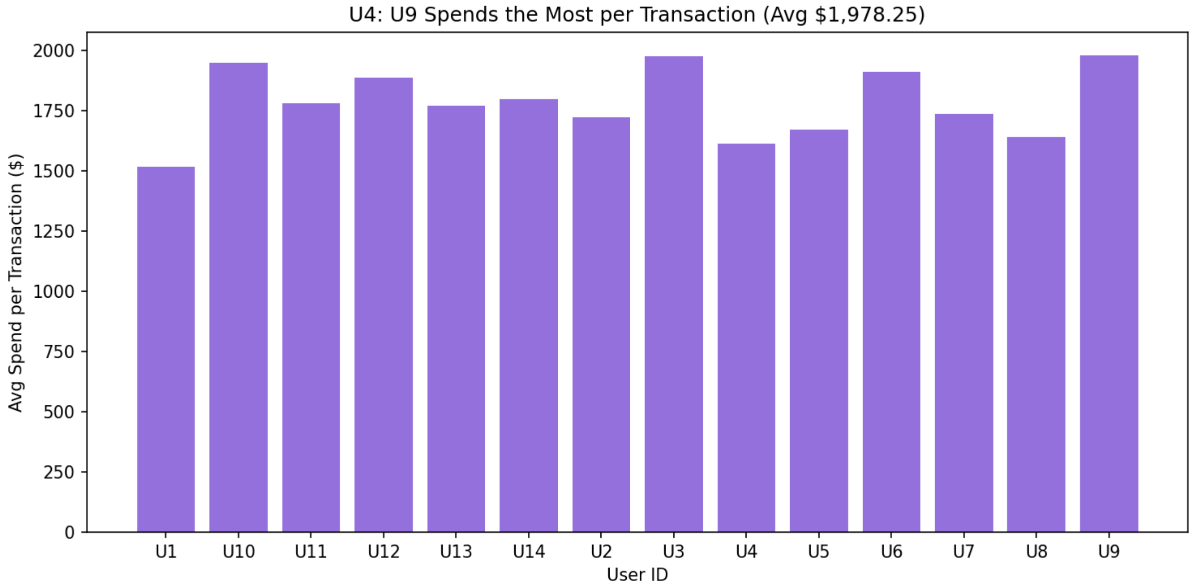
U12 also leads in total spending at $50,917.21, which is significantly higher than U13 who spent the least at $19,471.05. This is consistent with U12 having the highest transaction count. U6 ($40,102.65) and U9 ($39,564.99) follow closely behind, indicating they are high-value customers despite slightly fewer transactions.

U3. User Loyalty (Active Days)



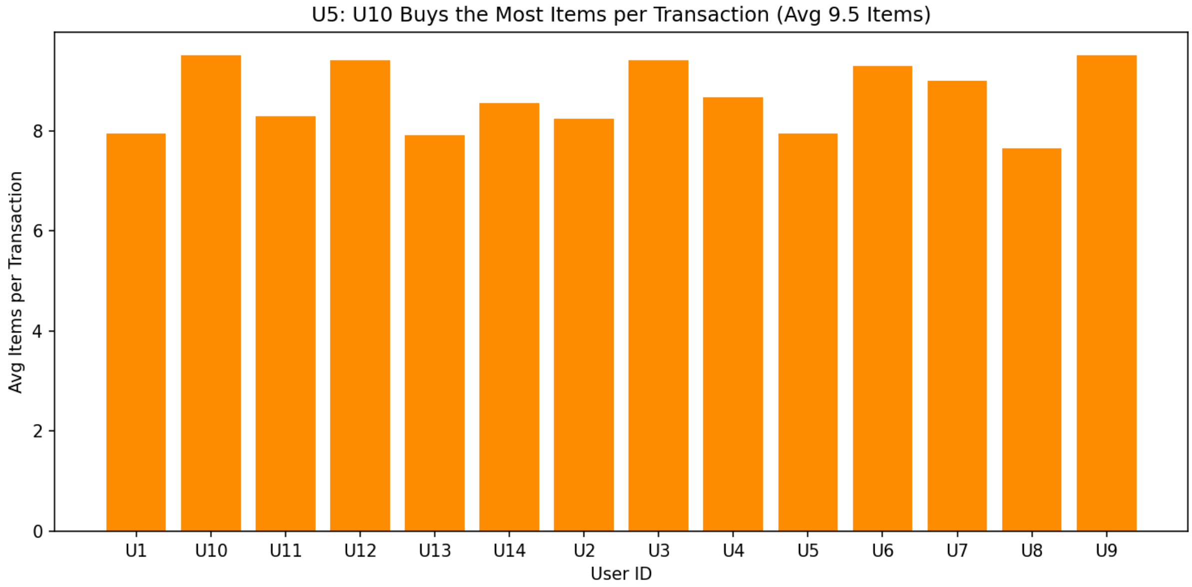
U11 and U12 both appear on 17 out of 30 days, making them the most consistently present users. U13 was active on only 8 days, reinforcing the pattern seen in transaction frequency. Importantly, some users appear multiple times on the same day, so active days measure loyalty separately from raw transaction count.

U4. Average Spend per Transaction per User



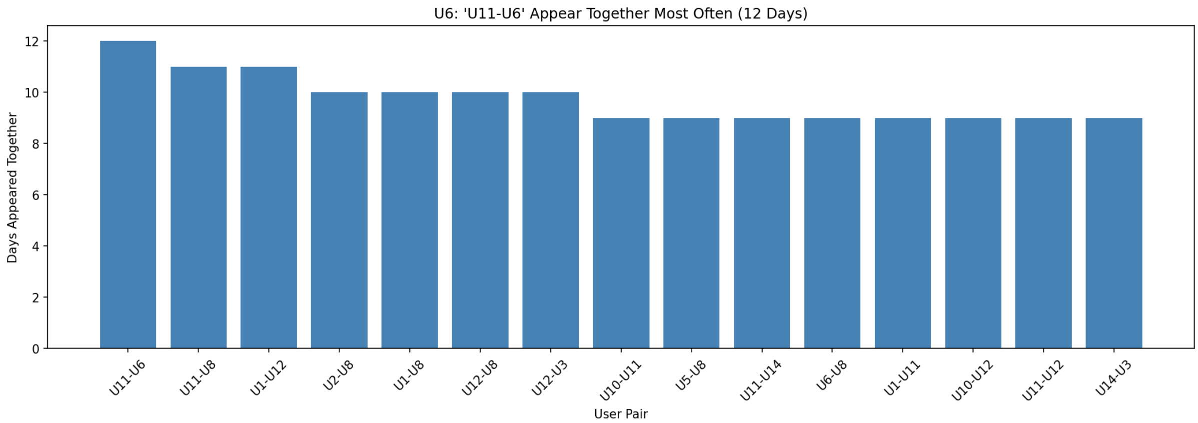
U9 and U3 have the highest average spend per transaction at $1,978.25 and $1,975.76 respectively. U1 has the lowest at $1,518.94. This reveals that some users tend to buy higher-priced items or fill larger baskets in each visit, even if they do not transact as often as U12.

U5. Average Basket Size per User



U10 and U9 both average 9.5 items per transaction. U8 and U13 have the lowest averages at 7.65 and 7.91 items respectively. A larger basket does not always mean higher spend since item prices vary, but users with the highest basket sizes consistently pick up more items regardless of the transaction.

U6. User Co-occurrence (Users Active on the Same Day)

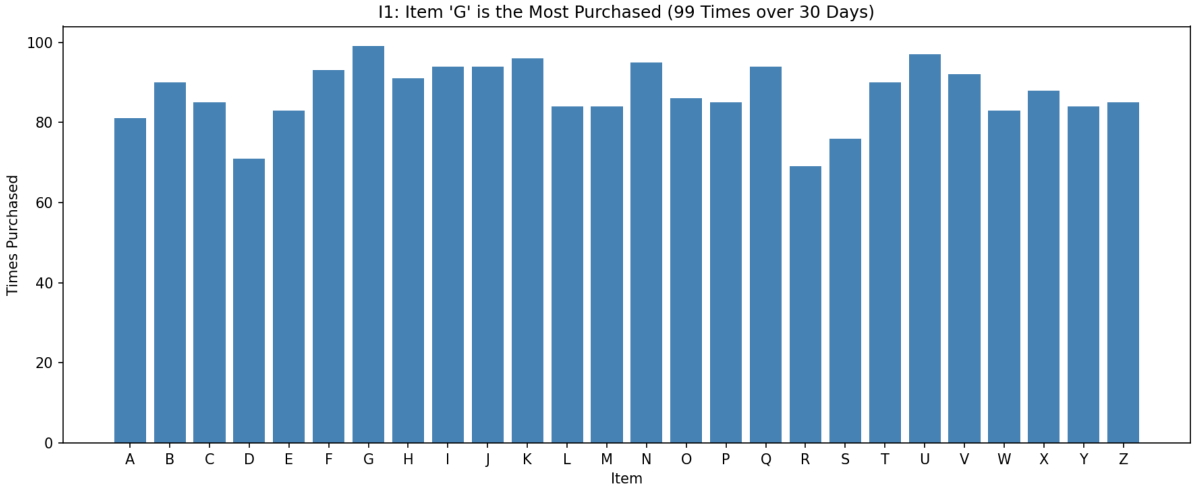


The pair U11 and U6 appeared together on 12 out of 30 days, making them the most frequently co-occurring users. U11 and U8 as well as U1 and U12 follow closely with 11 shared days each. Several other pairs co-occurred on 9 or 10 days, indicating a broad pattern of overlapping shopping days across the user base. This can reveal whether certain users shop together regularly, perhaps as couples, housemates, or people from the same area. The top 15 user pairs are shown above, ranked by the number of days they were both active.

2. Item Analysis

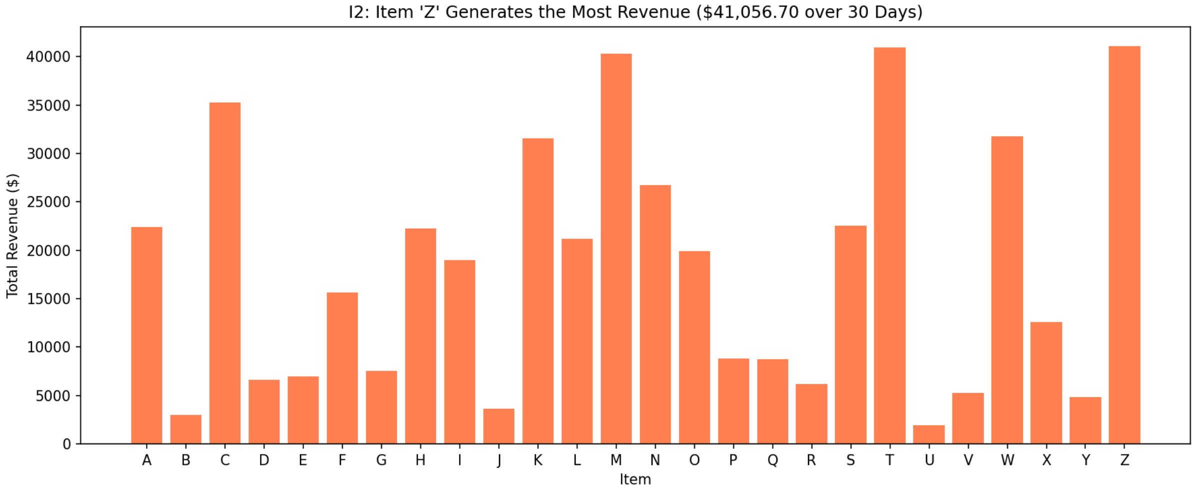
This section looks at which items are bought most, which generate the most revenue, which items tend to appear together, and which items show up consistently day after day.

I1. Item Purchase Frequency



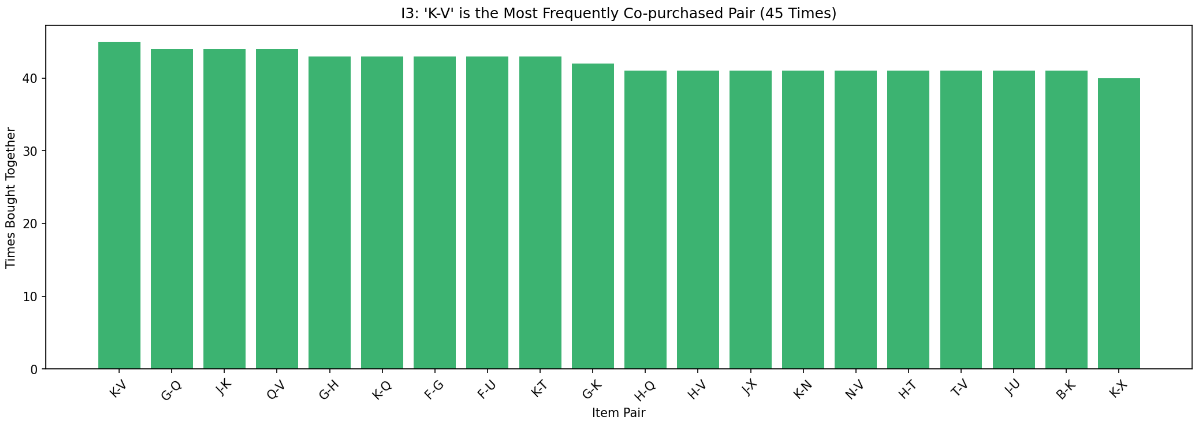
Item G was purchased 99 times across 30 days, the highest of any item. Item R was the least purchased at 69 times. The overall spread is relatively narrow, with most items landing between 84 and 99 purchases. This suggests no single item dominates demand heavily, though G has a clear edge.

I2. Item Revenue Contribution



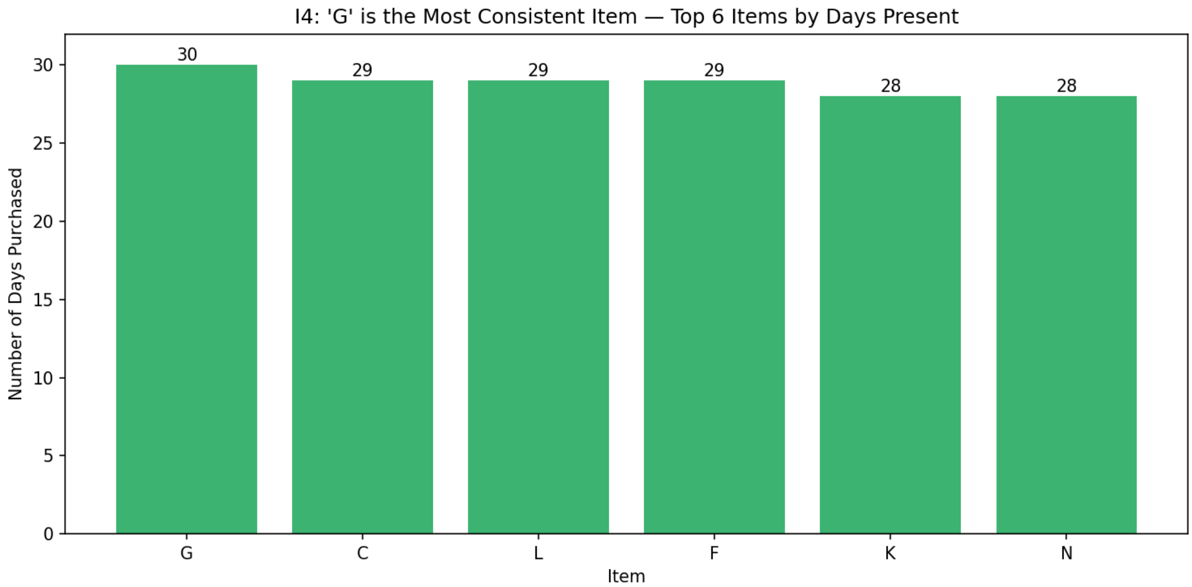
Item Z generated the most revenue at $41,056.70, followed closely by T ($40,957.20) and M ($40,268.76). Item U contributed the least at $1,945.82 despite being purchased 97 times. This is directly explained by price: U costs only $20.06 per unit while Z costs $483.02. High frequency does not guarantee high revenue when the item price is low.

I3. Item Co-occurrence (Bought Together)



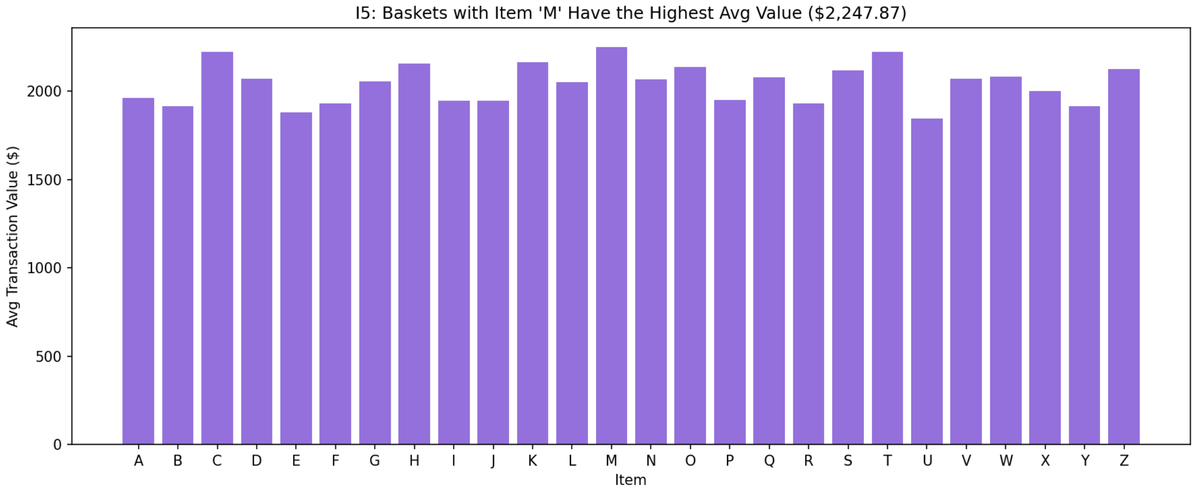
The pair K and V appeared together in 45 transactions, making them the most frequently co-purchased items. Several other pairs appeared 40 or more times. This kind of pattern is useful for understanding which items customers naturally group together when shopping, which can inform product placement decisions.

I4. Item Day Coverage (Daily Consistency)



Item G is the only item purchased on all 30 days. Items C, L, and F appeared on 29 days, while K and N appeared on 28 days. Item D was the least consistent, appearing on only 22 days. G stands out as the one item that every day saw some user pick up, regardless of how busy or slow the day was.

I5. Average Transaction Value When Item Is Present

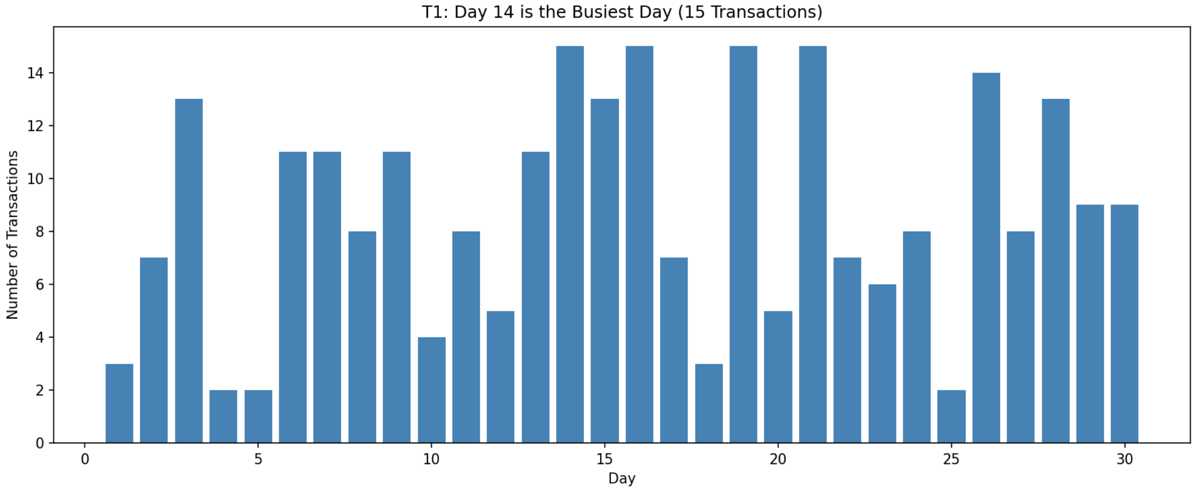


Transactions that include item M have the highest average total value at $2,247.87. Transactions with item U present have the lowest average at $1,845.74. This suggests that M tends to appear in larger, more expensive baskets while U tends to appear in smaller or lower-priced purchases.

3. Transaction Analysis

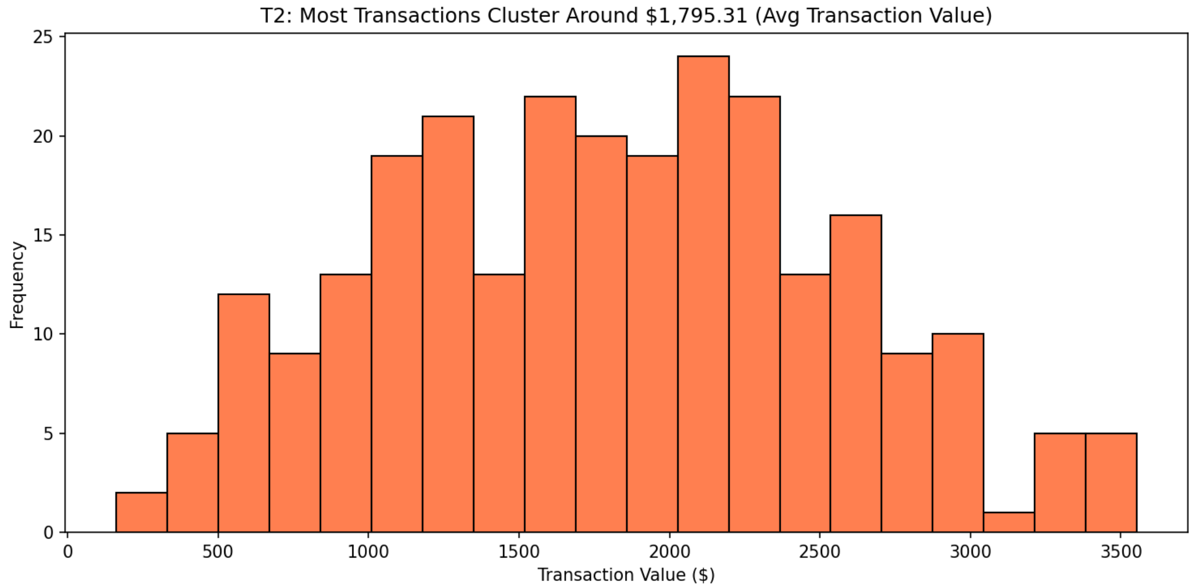
This section focuses on the transactions themselves: how many happened each day, what values they took, and how spending patterns shifted over the 30-day period.

T1. Transaction Volume per Day



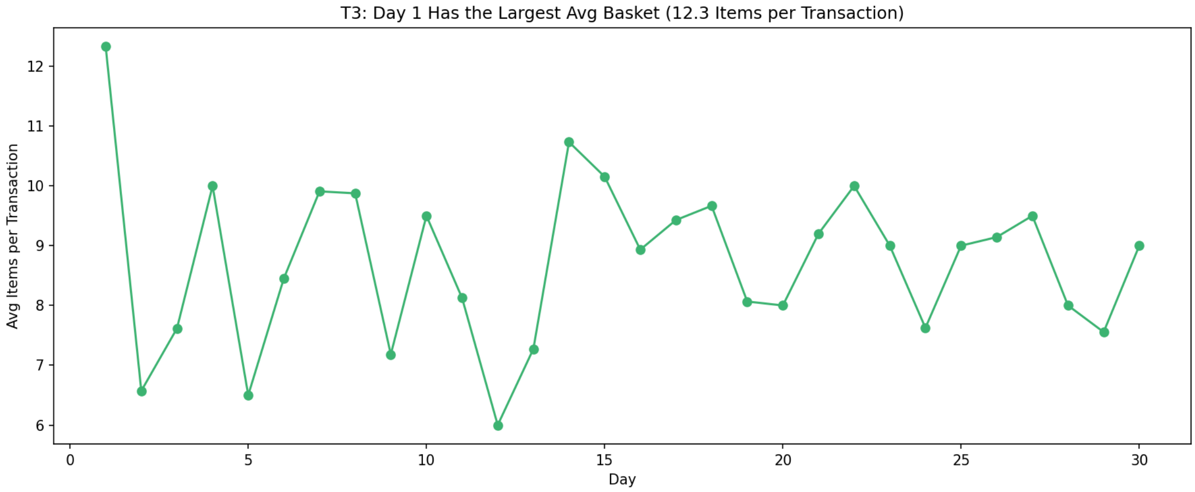
Day 14, Day 16, Day 19, and Day 21 each saw 15 transactions, the maximum recorded on any single day. Days 4, 5, and 25 had only 2 transactions each. The variation is large and does not follow a simple weekly pattern, suggesting demand fluctuations driven by other factors beyond just the day of the month.

T2. Distribution of Transaction Values



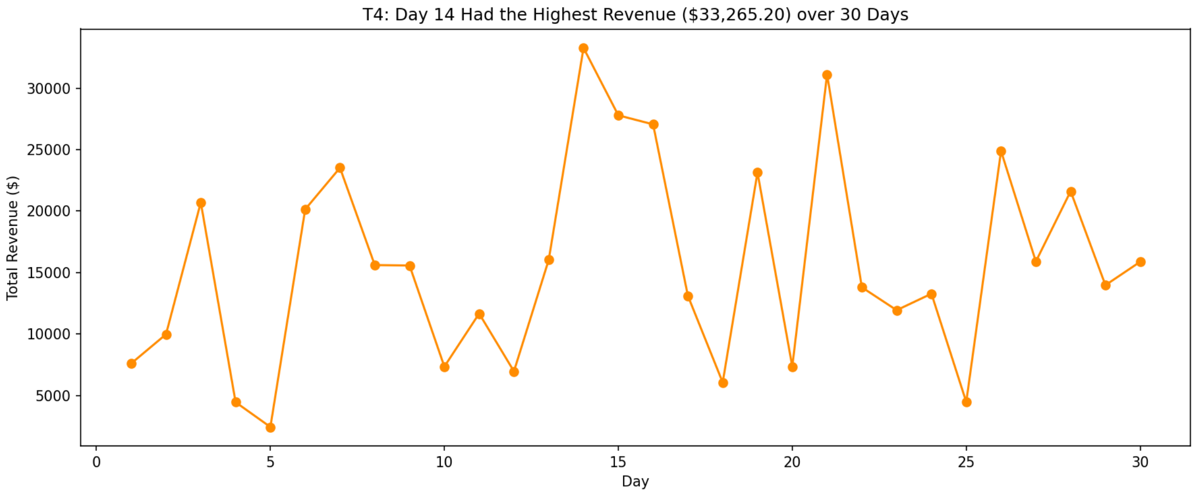
The average transaction value across all 260 transactions is $1,795.31. The distribution is spread between $162.74 and $3,553.00. Most transactions cluster in the $1,000 to $2,500 range. Very low value transactions tend to involve only a few cheap items, while high value ones include multiple expensive items.

T3. Average Basket Size per Day



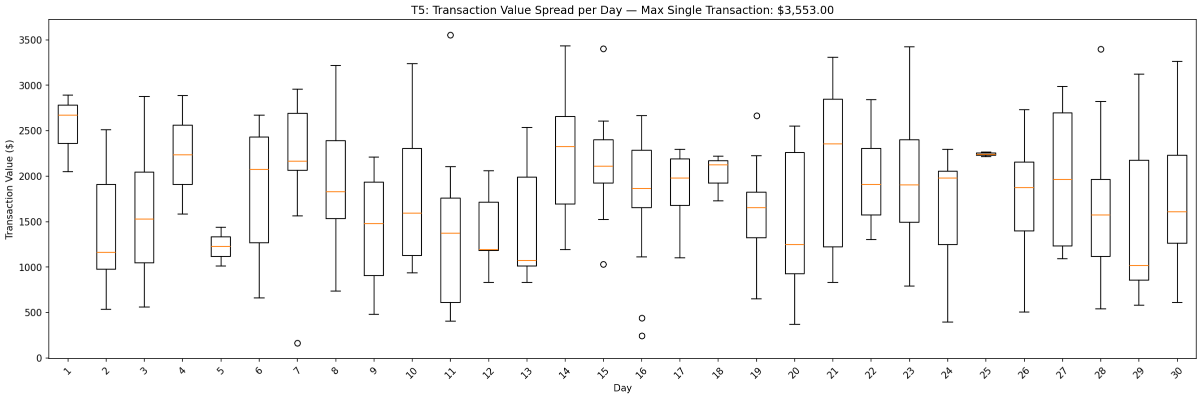
Day 1 had the largest average basket at 12.33 items per transaction, though it only had 3 transactions total. Days with more transactions tend to show lower average basket sizes, around 7 to 9 items. This could reflect that on quieter days, individual users tend to make larger single purchases.

T4. Daily Revenue Trend



Day 14 produced the highest single-day revenue at $33,265.20, coinciding with its 15 transactions. Day 5 produced the lowest revenue at $2,449.72. Revenue across the 30 days is irregular rather than trending steadily upward or downward. Total revenue over the period was $466,780.93.

T5. Transaction Value Spread per Day (Box Plot)



The box plot reveals that most days have a moderate spread of transaction values, but some days show wide variation. Days 4, 5, and 25 have very narrow boxes because they had very few transactions. Days with 13 to 15 transactions show broader distributions, with some transactions reaching above $3,000. The highest single transaction recorded was $3,553.00.